

WEEK — 01

IoT, Embedded Systems & Robotics

CNC · Microcontrollers · Sensors · Actuators · Projects



IoT



Embedded



Robotics



CNC



Sensors



Actuators

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IoT(Internet of Things)

Section 1.1.1 — Definition & Overview

What is IoT?

The Internet of Things (IoT) refers to a network of physical objects — "things" — embedded with sensors, software, and connectivity that enable them to collect and exchange data over the internet, without requiring human-to-human or human-to-computer interaction.

Key Areas of IoT:

- Smart Home
- Industrial IoT (IIoT)
- Healthcare & Wearables
- Smart Agriculture
- Smart Cities
- Retail & Supply Chain

1.1.2 Applications:

Internet of Things — Real-World Use Cases



Smart Home

Automated lighting, security cameras, thermostats, voice assistants

Smart Agri

Soil monitoring, drone irrigation, smart greenhouses

Industrial IoT

Predictive maintenance, asset tracking, process automation

Smart Cities

Traffic management, waste management, smart grids

Healthcare

Patient monitoring, smart hospitals, wearable health trackers

Retail & Supply

Inventory tracking, automated checkouts, cold chain

Embedded System

Section 1.2.1 — Definition & Overview

What is Embedded System?

An embedded system is a small computer system designed to perform a specific task inside a larger system. It consists of a processor, memory, and input/output units. Embedded systems are used in devices with size, cost, power, or weight limitations.

1.2.2 Applications:

Embedded Systems — Use Cases

Applications of Embedded Systems



GPS



Medical devices



Automated fare
collection



Home entertainment



Manufacturing



Electric vehicle
charging stations



Self-service
kiosks



GPS Systems — Navigation devices with real-time positioning

Medical Devices — Pacemakers, insulin pumps, diagnostic tools

Automotive Systems — ABS, ECU, ADAS, infotainment

Automated Fare Collection — Smart card ticketing systems

Home Entertainment — Smart TVs, set-top boxes, gaming consoles

ATM Machines — Cash dispensing and secure transactions

Manufacturing & Robotics — Industrial automation and control

EV Charging Stations — Battery management and billing

Self-Service Kiosks — Check-in, ordering, and ticketing

Robotics

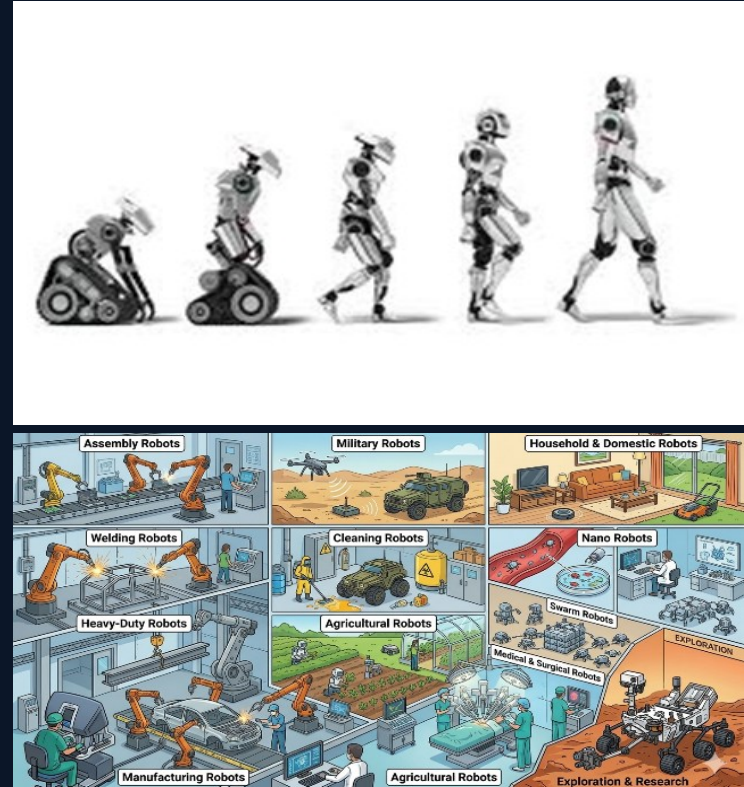
Section 1.3.1 — Definition & Overview

What is Robotics?

Robotics is the branch of engineering that involves the design, construction, operation, and use of robots. These are machines capable of carrying out complex tasks automatically. Robots sense their environment, process information, and perform physical actions.

Types of Robots:

- Assembly Robots
- Welding Robots
- Manufacturing Robots
- Military Robots
- Cleaning Robots
- Agricultural Robots
- Household & Domestic Robots
- Nano Robots
- Swarm Robots



1.3.2 Applications:

Robotics — Use Cases Across Industries



Assembly Robots

Used in industries for assembling products automatically.



Welding Robots

Used for seam welding and other industrial welding tasks.



Heavy-Duty Robots

Used for lifting heavy materials and industrial operations.



Manufacturing Robots

Used in factories and automobile industries for fast and accurate production.



Military Robots

Used for bomb detection, surveillance, and dangerous military operations.



Cleaning Robots

Used for cleaning toxic waste and hazardous industrial areas.



Agricultural Robots

Used for farming, planting, harvesting, and monitoring crops.



Household & Domestic Robots

Used for home cleaning and daily household tasks.



Nano Robots

Tiny robots used in medical and scientific applications.



Swarm Robots

Groups of robots working together to complete tasks efficiently.

Difference between IoT, Embedded System, and Robotics

1.4 Comparative Overview

Feature	IoT	Embedded System	Robotics
Definition	Network of connected objects	Dedicated computer in a product	Machines that perform tasks automatically
Main Focus	Connectivity & data exchange	Specific task execution	Physical action & motion
Internet	Essential 	Not required 	Optional 
Physical Action	Indirect	Often limited	Core requirement 
Human Interact.	Minimal	Minimal	Can be autonomous
Examples	Smart thermostat, GPS tracker	Washing machine, ABS	Surgical robot, drone, arm
Uses Embedded?	Yes — as backbone	Is the core	Yes — embedded inside

What is CNC?

Computer Numerical Control — Section 2

CNC (Computer Numerical Control) is the automated control of machining tools — such as drills, mills, lathes, routers, and laser cutters — using pre-programmed software and numerical codes. Rather than manual operation, a computer reads instructions and precisely moves cutting tools across multiple axes (X, Y, Z) to shape raw material into finished parts.

Key Advantages:

- Precision — sub-millimeter accuracy
- Repeatability — identical parts every run
- Speed — faster than manual machining
- Complexity — intricate shapes impossible by hand
- Safety — reduces operator exposure to danger

Types of CNC Machine



CNC Milling



Swiss-type CNC Lathe



CNC Lathe



Flat-bed CNC Lathe



Slant-bed CNC Lathe



CNC Pipe Threading Lathe

Role of CNC Machines in PCB Fabrication

How CNC is used in circuit board manufacturing



Drilling Via Holes

CNC drill machines bore hundreds of tiny holes (vias, through-holes) into PCB blanks with $\pm 0.05\text{mm}$ or better positional accuracy — impossible to achieve by hand.



Copper Trace Milling

CNC milling machines remove copper layers to create isolation channels and traces, making chemical-free single-layer circuit boards (ideal for prototyping).



Board Routing & Depaneling

CNC routers cut individual PCBs from large panels along exact outlines defined in Gerber/CAM files, producing clean edges and consistent board dimensions.



Speed & Quality Control

CNC automation dramatically increases throughput, reduces human error, ensures batch-to-batch consistency, and enables rapid design iteration.

Basic Idea of G-code

The language of CNC machines

G-code (Geometric Code) is the most widely used numerical control (NC) programming language. It tells a CNC machine what to do — where to move, how fast to move, and along what path — using simple alphanumeric commands.

G-code Building Blocks:

G words	Preparatory commands that set motion modes (e.g., G00 rapid move, G01 linear cut, G02 arc CW)
M words	Miscellaneous machine functions — spindle on/off, coolant, program stop/end
F word	Feed rate — controls how fast the tool moves during cutting operations
T word	Tool selection — specifies which tool from the magazine to use
S word	Spindle speed — sets RPM of the cutting spindle

G-Code Reference List

G Code	Function	G Code	Function
G00	Positioning at rapid travel;	G58	Set Datum;
G01	Linear interpolation using a feed rate;	G59	Set Datum;
G02	Circular interpolation clockwise;	G70	Finish cycle (Lathe);
G03	Circular interpolation, counterclockwise;	G71	Rough turning cycle (Lathe);
G04	Dwell	G72	Rough facing cycle (Lathe);
G17	Select X-Y plane;	G73	Chip break drilling cycle;
G18	Select Z-X plane;	G74	Left hand tapping (Mill);
G19	Select Z-Y plane;	G74	Face grooving cycle;
G20	Imperial units;	G75	OD groove pecking cycle (Lathe);
G21	Metric units;	G76	Boring cycle (Mill);
G27	Reference return check;	G76	Screw cutting cycle (Lathe);
G28	Automatic return through reference point;	G80	Cancel cycles;
G29	Move to a location through reference point;	G81	Drill cycle;
G31	Skip function;	G82	Drill cycle with dwell;
G32	Thread cutting operation on a Lathe;	G83	Peck drilling cycle;
G33	Thread cutting operation on a Mill;	G84	Tapping cycle;
G40	Cancel cutter compensation;	G85	Bore in, bore out;
G41	Cutter compensation left;	G86	Bore in, rapid out;
G42	Cutter compensation right;	G87	Back boring cycle;
G43	Tool length compensation;	G90	Absolute programming;
G44	Tool length compensation;	G91	Incremental programming;
G50	Set coordinate system (Mill);	G92	Reposition origin point (Mill);
G50	Maximum RPM (Lathe);	G92	Screw thread cutting cycle (Lathe);
G52	Local coordinate system setting;	G94	Per minute feed;
G53	Machine coordinate system setting;	G95	Per revolution feed;
G54	Set Datum;	G96	Constant surface speed (Lathe);
G55	Set Datum;	G97	Constant surface speed cancel;
G56	Set Datum;	G98	Feed per minute (Lathe);
G57	Set Datum;	G99	Feed per revolution (Lathe);

What do (G00, G01, G02, G03, G90, G91, G92, F, M03, M04, M05, T, M06, M08, M09, M00, M30) these mean in G-code?)

G-code Command Reference

Code	Name	Meaning
G00	Rapid Positioning	Moves tool at max speed — no cutting
G01	Linear Interpolation	Straight-line cut at controlled feed rate (F)
G02	Circular Interp. (CW)	Clockwise circular arc at feed rate
G03	Circular Interp. (CCW)	Counter-clockwise arc at feed rate
G90	Absolute Positioning	Coordinates measured from fixed origin
G91	Incremental Posit.	Coordinates relative to current position
G92	Set Coordinate System	Redefines current position as new origin
F	Feed Rate	Speed of tool movement during cutting (mm/min)
M03	Spindle ON (CW)	Starts spindle motor clockwise at set RPM

Code	Name	Meaning
M04	Spindle ON (CCW)	Starts spindle motor counter-clockwise
M05	Spindle OFF	Stops the spindle motor
T	Tool Select	Selects tool from magazine (e.g. T01 = tool #1)
M06	Tool Change	Executes the physical tool swap in spindle
M08	Coolant ON	Activates coolant pump/spray
M09	Coolant OFF	Deactivates coolant system
M00	Program Stop	Pauses program; waits for operator input
M30	End & Reset	Ends program, rewinds to beginning

Why microcontrollers are used?

Applications & Motivation



Low Cost

MCUs are extremely affordable (pennies to a few dollars), integrating CPU, memory, and I/O on one chip, removing the need for multiple discrete components.



Low Power Consumption

Designed for efficient operation, MCUs can run for years on a battery in sleep modes — ideal for IoT sensors, wearables, and remote devices.



Task-Specific Efficiency

No OS overhead, no unused processing power. Real-time deterministic response is predictable and fast for the dedicated task.



Compact & Integrated

Entire control systems fit on a fingernail-sized chip, reducing PCB area, weight, and assembly complexity.



Ease of Programming

Supported by C/C++, MicroPython, Arduino IDE, and visual tools, lowering the barrier for rapid prototyping and development.



Broad Application Fields

Consumer Electronics · Automotive Industry · Medical Devices · Industrial Automation · Communication Systems

A list of available microcontrollers in Bangladesh

With approximate prices (BDT) — 2025

Microcontroller / Development Board	Approximate Price (BDT)
<u>Arduino Boards</u>	
Arduino Uno R3	949
Arduino Nano (ATmega328)	363
Arduino Pro Mini 328 (5V/16MHz)	430
Arduino UNO R4 Minima 32bit 48MHz	1,650
Arduino Pro Micro Atmega32U4 5V/16MHz	680

<u>ESP & Wireless Microcontrollers</u>	
ESP32 NodeMCU Development Board Wireless Wifi	427
ESP8266 NodeMCU Lua V3	350
ESP32-CAM (with Camera Module)	940
ESP32-S3 Microcontroller 2.4 GHz Wi-Fi Development Board	1,590
ESP32 C6 DevKit-C N16R8	950
Wemos D1 Mini ESP8266	320

A list of available microcontrollers in Bangladesh

With approximate prices (BDT) — 2025

<u>STM32 & Advanced ARM Boards</u>	
<u>STM32F103C8T6 "Blue Pill" Board</u>	480
<u>STM32G431CBU6 Mini Core Development Board</u>	1,550
<u>STM32F401RCT6 MicroPython Compatible</u>	974
<u>STM32F411CEU6 Black Pill Development Board</u>	874
<u>STM8S103F3P6 STM8 Blue Core</u>	350
<u>Raspberry Pi & Teensy</u>	
<u>Raspberry Pi Pico 2 (RP2350)</u>	1,250

<u>Raspberry RP2040 Core Board 16MB</u>	1,450
<u>Teensy 4.1 Development Board</u>	8,945
<u>Bare AVR, PIC & RISC-V Chips</u>	
<u>ATmega32A</u>	430
<u>ATtiny15L</u>	270
<u>PIC18F4550</u>	762
<u>PIC16F877A</u>	2,250
<u>PIC16F628</u>	135
<u>CH32V003 Development Board 48Mhz RISC-V</u>	390

What are sensors and actuators?

Section 4 — Input & Output Devices



SENSORS

Sensors are input devices that detect physical or chemical changes in the environment (temperature, light, motion, gas, sound, distance) and convert them into electrical signals a microcontroller can read and process.

Signal Flow:

Environment → Physical Quantity → Electrical Signal → MCU → Decision

Used for:

- Data acquisition
- Automation & control
- Safety & protection
- Decision-making systems



ACTUATORS

Actuators are output devices that convert electrical control signals from a microcontroller into physical action — movement, heat, light, or sound. They are the 'muscles' of an embedded system.

Signal Flow:

MCU Decision → Electrical Signal → Actuator → Physical Motion/Action

Used for:

- Motion & movement control
- Switching high-power devices
- Valve & flow control
- Sound & light output

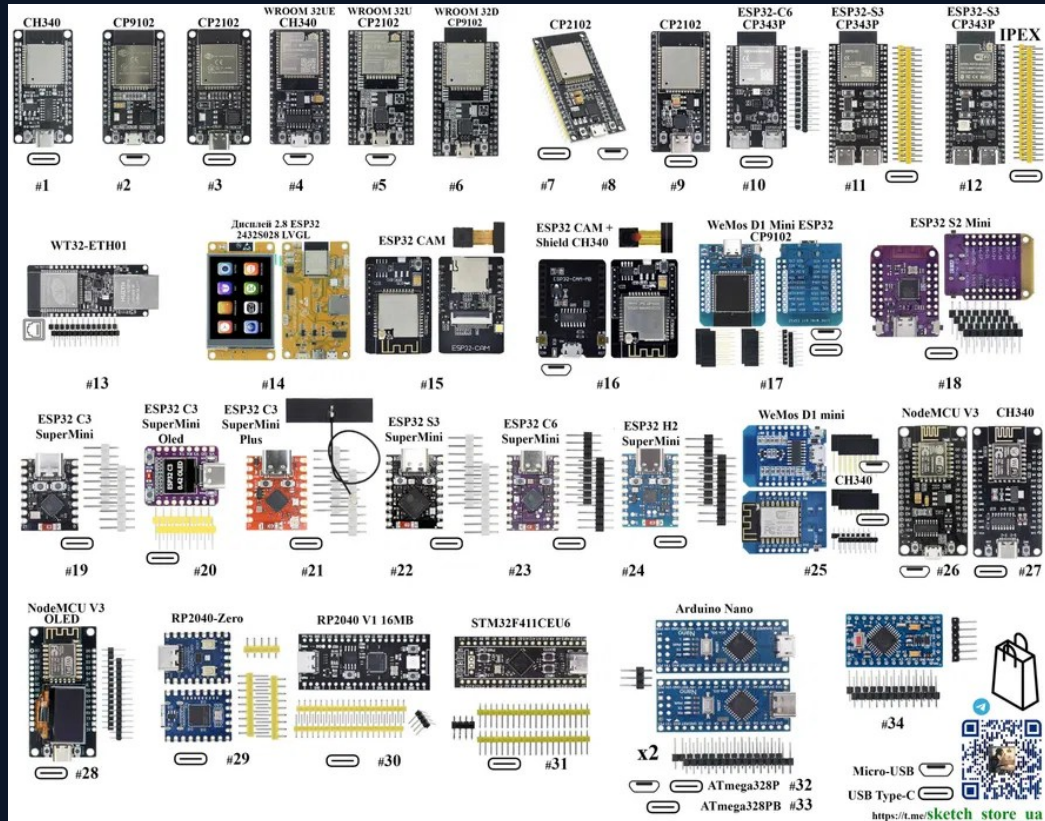
Functionalities of common sensors and their applications

Temperature, Gas, Light, Motion, Sound, Ultrasonic, IR & more

	Sensor	Type	How It Works	Applications
	DHT11 / DHT22	Temp & Humidity	Measures ambient temperature and humidity via capacitive/resistive sensing	Smart home, HVAC, weather stations
	HC-SR04	Ultrasonic	Emits 40kHz sound pulses; measures echo return time to calculate distance	Obstacle avoidance, tank level, parking
	MQ Series Gas	Gas / Air Quality	Resistance changes with target gas contact on heated metal oxide surface	Fire detection, CO alarm, air quality IoT
	LDR	Light	LDR changes resistance with light; BH1750 gives digital lux via I2C	Auto lighting, solar trackers, displays
	PIR	Motion	Detects IR radiation difference from warm bodies across two pyroelectric zones	Security alarm, auto-lighting, occupancy
	Sound KY-037	Sound	Microphone converts pressure waves to voltage; comparator outputs digital HIGH	Clap detection, noise monitoring, voice
	IR FC-51	Infrared	IR LED emits light; phototransistor detects reflection for proximity/line following	Line following robots, remote control decode
	Soil Moisture	Soil Water	Capacitive/resistive sensor measures soil water content as analog voltage	Smart irrigation, plant monitoring, agri-IoT
	Potentiometer	Rotary Input	Variable resistor; analog voltage proportional to rotational position	Volume control, angle sensing, user input
	4x4 Keypad	User Input	Matrix of 16 buttons; MCU scans rows/columns to detect key press	PIN entry, menu navigation, password systems

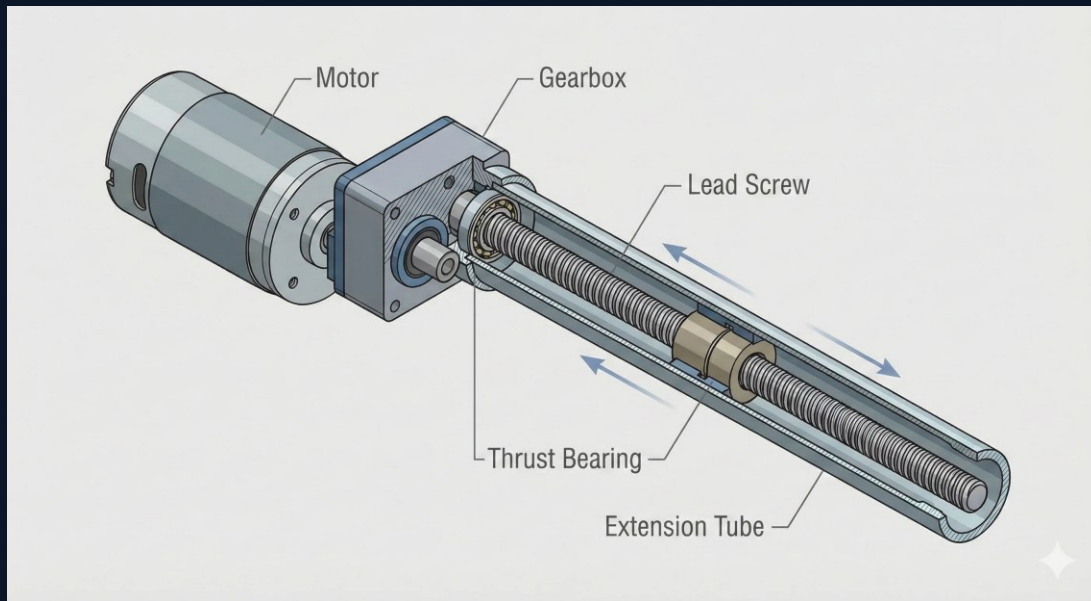
Different Types of Sensors

Visual overview of sensor modules



Actuator Structure

Motor · Gearbox · Lead Screw · Thrust Bearing · Extension Tube

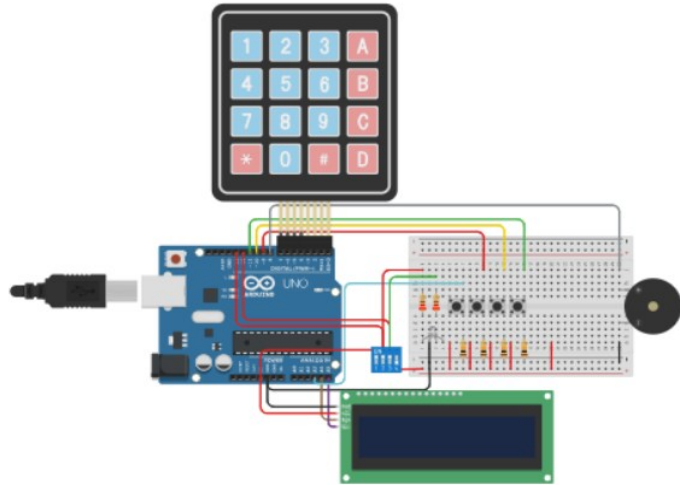


Motor	Gearbox	Lead Screw	Thrust Bearing	Extension Tube
Converts electrical energy to rotational motion	Reduces speed, increases torque	Converts rotation to linear motion	Absorbs axial loads during extension	Translates along axis as output stroke

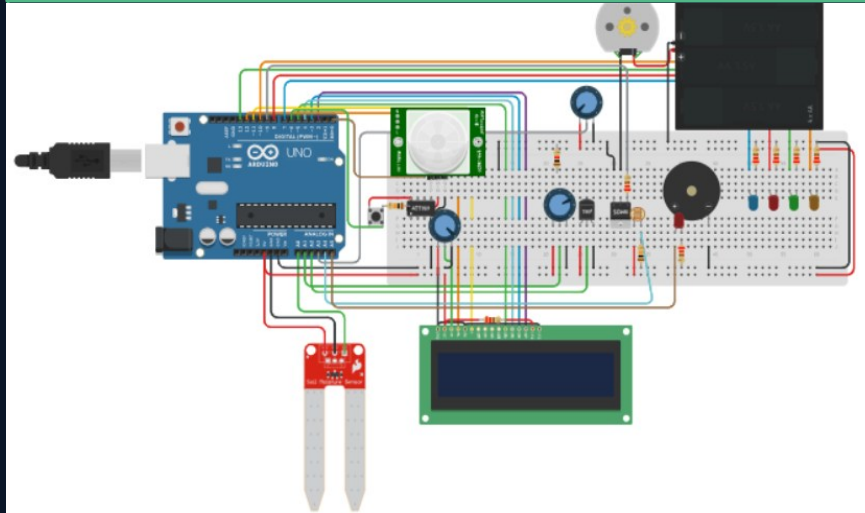
Real-Life System Example

Two integrated projects demonstrating microcontrollers, sensors, actuators, user input, control logic, data logging, and real-world problem solving.

 Advance Traffic Light System



 Smart Farm Automation



Advance Traffic Light System and Smart Farm Automation

Project Summaries & Repository Links



Advance Traffic Light System

- Arduino Uno as central controller
- 4×4 Keypad for password-protected config
- 16×2 LCD for status display
- RGB LED array for traffic signals
- Buzzer for pedestrian alert tones
- DIP switch for manual test mode
- Automatic traffic cycle with timers
- Password-protected configuration mode

GitHub Repository:

[View on GitHub →](#)



Smart Farm Automation

- Arduino Uno as central controller
- Soil moisture sensor for irrigation control
- DHT11 for temperature & humidity monitoring
- LDR for light intensity sensing
- PIR sensor for security & intruder detection
- Speaker for audio alerts on events
- LCD display for live readings
- Data logging and recovery features

GitHub Repository:

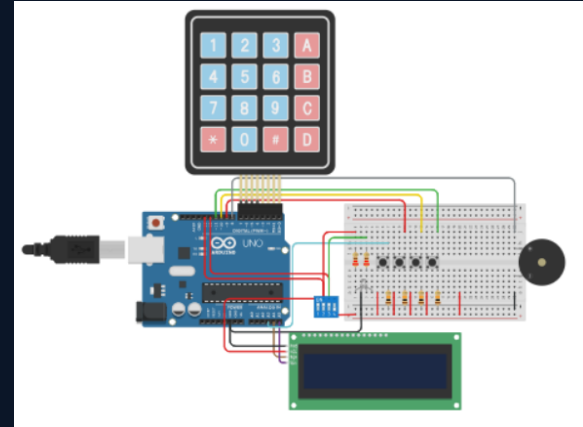
[View on GitHub →](#)

Advance Traffic Light System

TinkerCAD circuit simulation screenshot

Technical Challenges & Future Scope

A notable challenge in this advanced loop is handling blocking delays. Currently, exiting the automated cycle requires specific timing to interrupt the loop. Future iterations of this system could implement hardware interrupts or non-blocking timers (`millis()`) to allow instantaneous mode switching. Additionally, writing the administrator's custom delays to EEPROM would ensure the settings survive power cycles, and integrating an RTC (Real-Time Clock) could allow the system to switch to "night mode" automatically.



Components: Arduino Uno · 4x4 Keypad · 16x2 LCD · RGB LED · Buzzer · DIP Switch

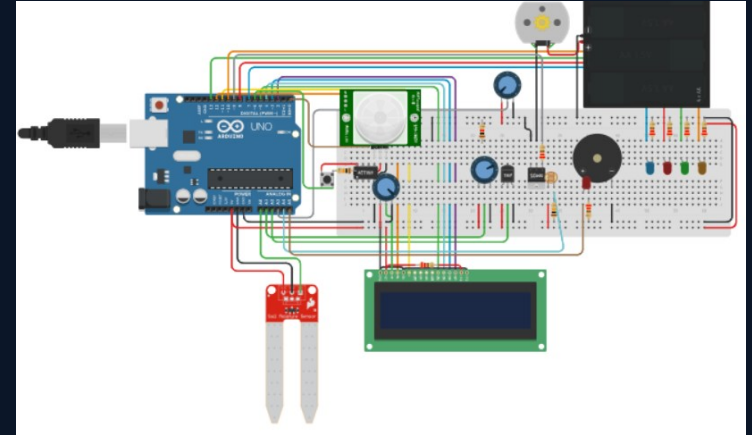
[View Repository →](#)

Advance Traffic Light System

TinkerCAD circuit simulation screenshot

Technical Challenges & Future Scope

Developing a dual-purpose system on a constrained microcontroller presented several technical hurdles. A primary challenge was managing **timing conflicts**; the system had to balance standard farming automation delays with the need for instantaneous, interrupt-like responsiveness for intruder logging. Furthermore, managing numerous analog and digital sensors alongside multiple actuators quickly exhausted the available I/O pins on the Arduino Uno, and continuous loop updates caused noticeable LCD flicker.



Components: Arduino Uno · Soil Moisture · DHT11 · LDR · PIR Sensor · Speaker · LCD Display

[View Repository →](#)

References

All source links — click to open

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TechShop BD

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GitHub — All Projects

<https://github.com/Preetam21022/TinkerCAD-Projects/tree/main>

GitHub — Traffic Light

<https://github.com/preetam21022/tinkercad-projects/tree/main/Advance%20Traffic%20Light%20System>

GitHub — Smart Farm

<https://github.com/Preetam21022/TinkerCAD-Projects/tree/main/Smart%20Farm%20Automation>

WEEK
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Thank You!

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